



EMT 3203: Solid Mechanics II

B.Sc. Mechatronics Engineering

Continuous Assessment Test I

2025

Instructions:

- ✪ Answer all questions. The test has a total of 30 marks.
- ✪ You have 1 hour to complete this test.
- ✪ Write your name, registration number, and your answers in the provided answer booklet.

Question 1

[6 marks]

A PCB in an electronics rack has length 180 mm, width 120 mm, and thickness 1.6 mm. The material is FR4 with Young's modulus $E = 24 \text{ GPa}$. The component weight of 10 N total is distributed uniformly across the board, and the support configuration is fixed-fixed at both ends. Calculate the maximum deflection and assess whether it meets the design standard (deflection $< L/100$).

Question 2

[12 marks]

A material handling robotic arm is modeled as a cantilever beam with length $L = 2.5 \text{ m}$. The arm experiences a distributed load of $w = 75 \text{ N/m}$ due to its self-weight and a point load $P = 1200 \text{ N}$ at the tip representing the payload. The beam has a rectangular aluminum cross-section of $80 \text{ mm} \times 10 \text{ mm}$ with second moment of area $I = 5.33 \times 10^6 \text{ mm}^4$ and distance from neutral axis $c = 40 \text{ mm}$. The material is aluminum alloy with yield strength $\sigma_{\text{yield}} = 275 \text{ MPa}$.

- (a) Calculate support reaction forces at the fixed base. [2 marks]
- (b) Construct the shear force diagram and identify the maximum shear force. [3 marks]
- (c) Construct the bending moment diagram and identify the critical location. [4 marks]
- (d) Calculate the maximum bending stress and assess structural adequacy with an appropriate safety factor. [3 marks]

Question 3

[12 marks]

An electric train pantograph arm maintains contact with overhead power lines. The arm has a pin joint at point A ($x = 0$) and total length $L = 1200$ mm. A spring is located 300 mm from pin joint A, and a wire contact force $P_1 = 800$ N acts downward at the tip. The arm is constructed from a hollow steel tube with outer diameter OD = 50 mm and wall thickness 4 mm, giving a second moment of area $I = 2.45 \times 10^6$ mm⁴ and distance from neutral axis $c = 25$ mm. The material is high-strength steel with yield strength $\sigma_{\text{yield}} = 250$ MPa, and a safety factor of 3.0 is required.

- (a) Calculate the required spring force using equilibrium equations. [2 marks]
- (b) Construct the shear force diagram showing all regions. [3 marks]
- (c) Determine the bending moment distribution and locate the maximum moment. [4 marks]
- (d) Calculate the maximum bending stress at the critical section and evaluate the safety factor to determine design adequacy for dynamic railway operations. [3 marks]